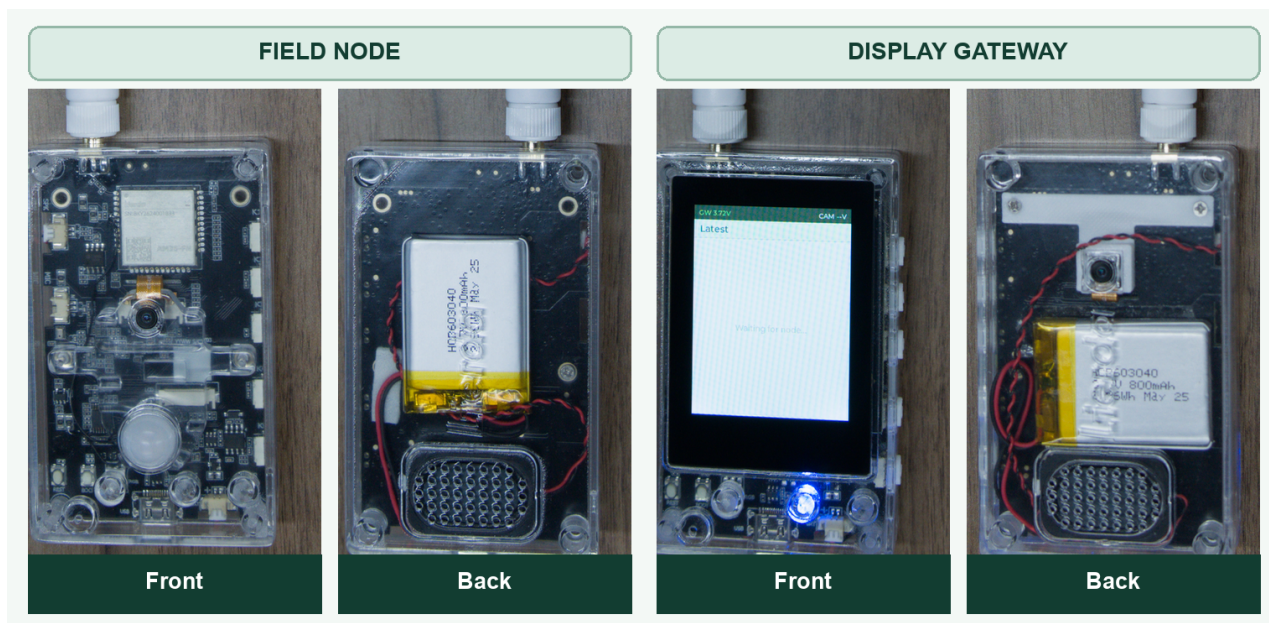


# AM36 WildCam

## Feature Guide



## Product Overview

**AM36 WildCam** is an open-source kit built on the AM36 hardware platform. It consists of a field camera and a handheld receiver. The field camera can be mounted on a tree, fence, farm, woodland site, or outdoor entrance and normally remains in a low-power state. A timer, PIR motion event, or sound event can trigger a capture, after which the image is returned to the handheld receiver over LR2021 FLRC.

The field camera integrates a camera, PIR motion sensor, and microphone. The handheld receiver provides an LCD, physical buttons, and an HTTP image gallery, displaying images, battery voltage, signal quality, and transfer rate. The system supports both LoRa and FLRC: normal operation uses FLRC directly, while low-power operation first wakes the camera with a LoRa long preamble and then switches to FLRC for image and optional audio transfer. Manual, timer, PIR, and sound-triggered captures can be viewed on the receiver, link-status page, and web gallery, demonstrating the ESP32 + LR2021 platform in long-range visual sensing applications.

# 01 Capabilities and Applications

From an outdoor trigger to image presentation, AM36 provides a complete and intuitive long-range wireless sensing path.

## Product Positioning

AM36 WildCam is a reference design for a long-range wireless imaging node. The field camera is deployed in woodland, on a farm or fence, or at an outdoor entrance, where it captures images periodically or in response to events. The display gateway receives and presents the images, reports link status, and provides access to the HTTP image gallery.

## Core Capabilities

|                                                                                                                                                |                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Long-Range Wireless</b><br>No on-site Wi-Fi coverage is required. LoRa provides low-power wake-up, while FLRC carries image and audio data. | <b>Low-Power Operation</b><br>PIR and remote wake-up remain available in low-power mode; FLRC starts only when service data must be transferred. |
| <b>Event Capture</b><br>The camera captures images, and the microphone can retain about five seconds of audio preceding the capture.           | <b>Local Visualization</b><br>The LCD shows reception progress, RSSI, rate, packet loss, transfer time, and the received image.                  |

## Typical Applications

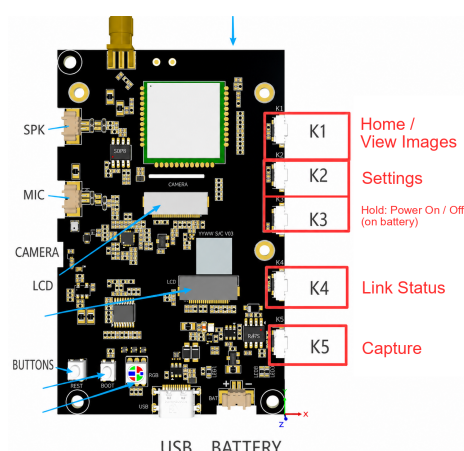
|                                                                                                                       |                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Wildlife Observation</b><br>Use timer or PIR captures in woodland, at bird activity sites, or along animal trails. | <b>Farm and Forestry Monitoring</b><br>Observe livestock, people, or unusual activity in orchards, fields, and woodland boundaries.          |
| <b>Outdoor Security</b><br>Capture events around fences, temporary work sites, equipment boxes, or campsites.         | <b>Wireless Demonstration</b><br>Trigger a live capture at an exhibition or customer demonstration and show image transfer and link quality. |

## End-to-End Workflow

|                                                                   |                                                                                     |                                                               |                                                                            |                                                                         |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>1</b><br><b>Trigger</b><br>Manual, timer, PIR, or sound event. | <b>2</b><br><b>Capture</b><br>The field camera obtains an image and optional audio. | <b>3</b><br><b>Transfer</b><br>Data returns over LR2021 FLRC. | <b>4</b><br><b>Present</b><br>The gateway shows the image and link status. | <b>5</b><br><b>Browse</b><br>The web gallery provides images and audio. |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------|

## 02 Gateway Interface and Buttons

Touch settings and physical buttons support on-site operation without an external device.



### Gateway Mode

| Button | Press                 | Function                                     |
|--------|-----------------------|----------------------------------------------|
| K1     | Short                 | Open the Images page.                        |
| K1     | Hold 1.5 s            | Switch from gateway mode to field-node mode. |
| K2     | Short                 | Open the Settings page.                      |
| K3     | Hold on battery power | Turn device power on or off.                 |
| K4     | Short                 | Open the Link Status page.                   |
| K5     | Short                 | Start a remote capture.                      |

### Field-Node Mode

| Button | Press                 | Function                                     |
|--------|-----------------------|----------------------------------------------|
| K2     | Short                 | Switch from field-node mode to gateway mode. |
| K3     | Hold on battery power | Turn device power on or off.                 |

Touch screen: tap controls directly. Swipe left or right to cycle through Images, Link Status, and Settings. Page navigation is temporarily ignored during an image transfer or while the transfer result is being displayed.

## 03 Capture Triggers

Four trigger types share the same image capture and return path and can be combined as required.

### Multiple Capture Triggers

Different events start the same image capture and transfer path.

#### Manual

##### Button / screen

K5 or CAPTURE  
on Settings page

#### Timer

##### Timer

Capture automatically  
at selected interval

#### PIR

##### PIR motion

Detect motion and  
start a capture

#### Sound

##### Sound

Sound exceeds the  
selected threshold

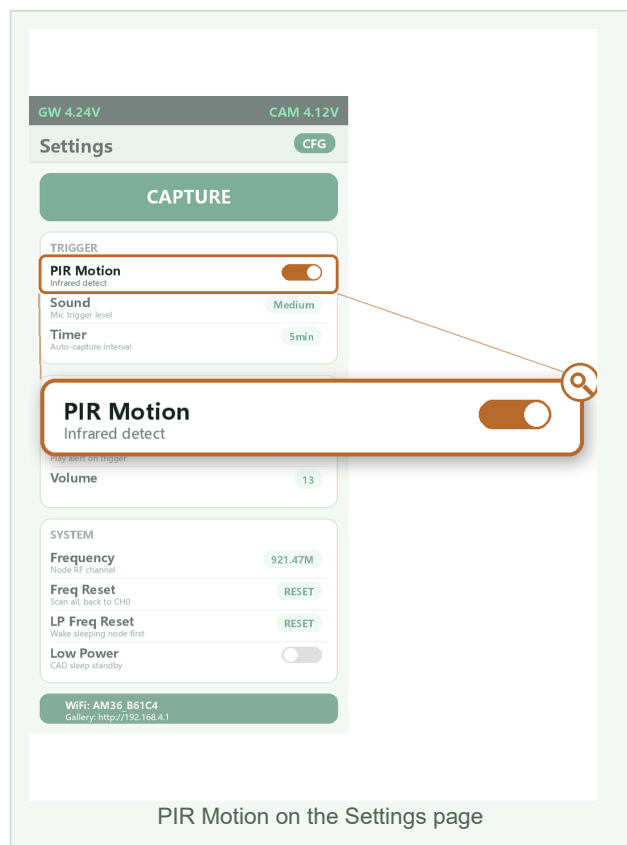
After a trigger: capture image -> wireless transfer -> gateway display and storage.

| Trigger           | Setup or Operation                                                  | Result                                                                                                     |
|-------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| On-Screen Capture | Tap CAPTURE at the top of the Settings page                         | The gateway starts a remote capture and opens the reception-progress page.                                 |
| Physical Button   | Press K5 on the display gateway                                     | A remote capture can start from any page; the UI switches to the reception page.                           |
| Timer             | Select Off, 10s, 30s, 1min, 5min, 10min, 15min, 20min, 30min, or 1h | The field camera captures and returns an image at the selected interval.                                   |
| PIR Motion        | Enable PIR Motion                                                   | A GPIO12 high level starts a capture and return; detection is re-enabled after 15 seconds.                 |
| Sound             | Select Low, Med, or High                                            | Audio above the selected threshold starts a capture after about 0.5 seconds to preserve the trigger sound. |

Manual capture starts with a gateway command. Timer, PIR, and sound events originate at the field camera. A new capture is not started while the capture task is busy.

## 04 PIR Motion Trigger

Automatically capture on an infrared motion event for birding, wildlife observation, and outdoor security.



### PIR Motion

When enabled, the field camera continuously monitors the PIR input. Detected motion starts an event capture and pushes the image to the display gateway.

### After a Trigger

The PIR input triggers on a GPIO12 high level. Detection is temporarily disabled after an event and re-enabled 15 seconds later to prevent repeated captures from a sustained high level.

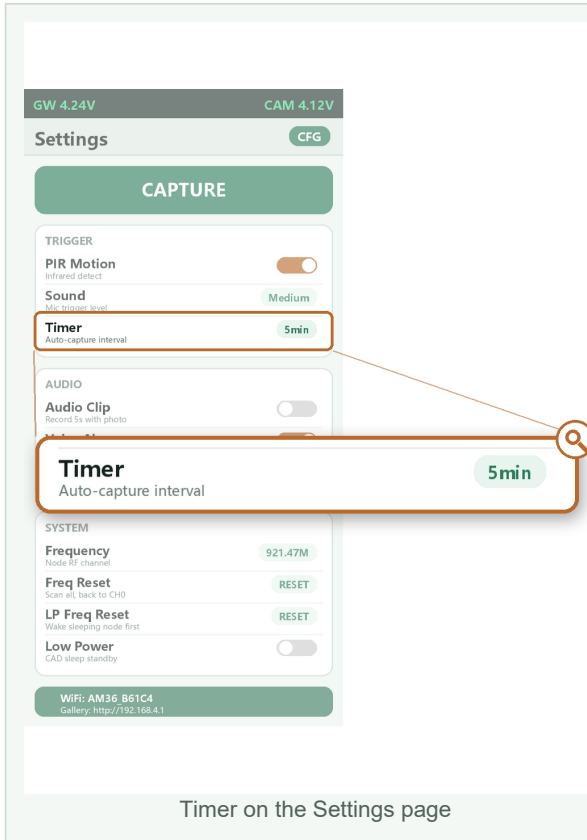
### Demonstration Use

The gateway can receive an image when a bird passes, an animal moves, or a person or object moves near a campsite.

PIR only detects motion and starts a capture. Audio Clip independently controls whether audio accompanies the image, while Voice Alarm controls the local alert after the event.

## 05 Timer Capture

Automatically record site changes at a fixed interval without continuously operating the gateway.



### Timer

After an interval is selected, the field camera automatically captures and returns an image at that interval. Select Off to disable timer capture.

### Available Intervals

Off, 10 seconds, 30 seconds, 1 minute, 5 minutes, 10 minutes, 15 minutes, 20 minutes, 30 minutes, and 1 hour.

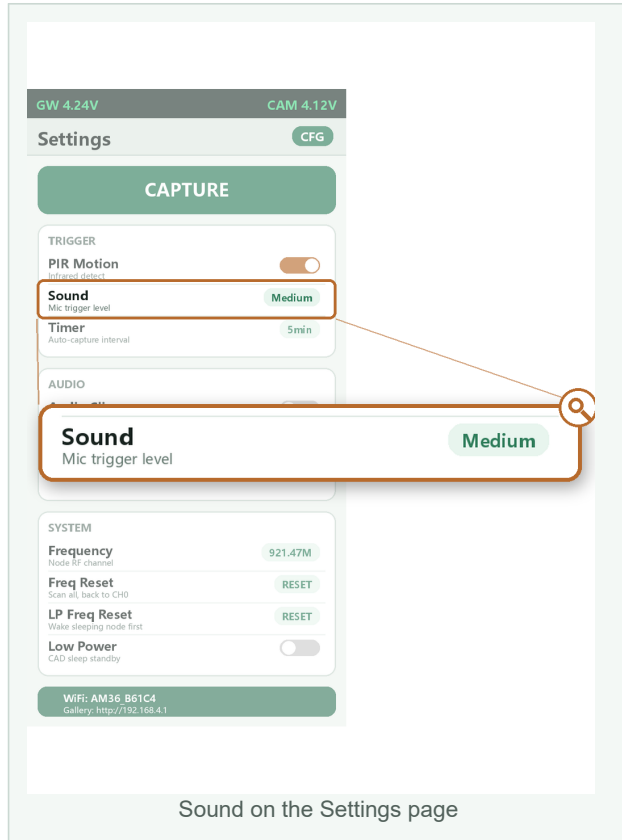
### Demonstration Use

Observe changes in bird activity, record a campsite environment, or collect site images at scheduled times.

Timer capture runs at the selected interval. Timer, PIR, and sound triggers are independent and can be enabled separately.

## 06 Sound Trigger

Automatically capture when sound exceeds the selected threshold. This trigger operates alongside PIR and timer capture.



### Sound

When Low, Med, or High is selected, the field camera continuously monitors microphone level. Sound above the corresponding threshold starts a capture and returns the image. Select Off to disable sound triggering.

### Sensitivity

Low is the most sensitive, Med provides medium sensitivity, and High requires a louder sound.

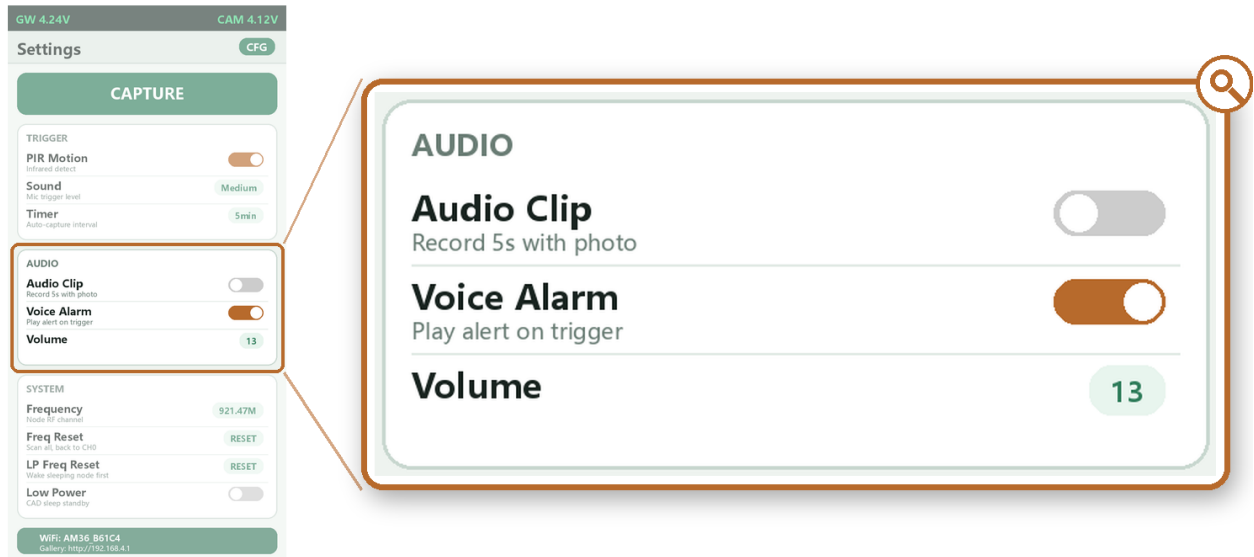
### After a Trigger

Capture starts after a delay of about 0.5 seconds to retain the trigger sound. Event triggers share a 15-second guard interval.

Sound controls only whether audio level starts a capture. Audio Clip independently controls whether recorded sound is returned with the image.

## 07 Audio Recording and Alerts

Audio Clip, Voice Alarm, and Volume independently control recording, local alerts, and gateway playback volume.



AUDIO group on the Settings page

| Setting     | Behavior                                                                                                                           | Result                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Audio Clip  | The field camera continuously retains about five seconds of pre-encoded audio and returns it with the image when a capture occurs. | The gateway plays it once after reception and stores it with the image; the web gallery shows an AUDIO label.   |
| Voice Alarm | After a PIR- or sound-triggered image has been captured and sent, the field camera plays the built-in alert.                       | Provides a local audible indication. Manual and timer captures do not play this alert.                          |
| Volume      | Sets gateway audio playback volume, cycling from 0 through 15.                                                                     | Controls playback of received Audio Clips. Voice Alarm plays at the project-defined volume on the field camera. |

Audio Clip controls whether sound is returned with the image. Voice Alarm controls whether the field camera plays an alert after an event. The two settings are independent.

## 08 RF Channel and Frequency Reset

Gateway and node share one RF channel; Settings changes it safely or recovers a lost node.

GW 4.24V
CAM 4.12V

**Tap again to reset**
CONFIRM

Volume
13

**SYSTEM**

**Frequency**  
Node RF channel
921.47M

**Freq Reset**  
Scan all, back to CH0
SURE?

**LP Freq Reset**  
Wake sleeping node first
RESET

**Low Power**  
CAD sleep standby
☐

**WiFi: AM36 B61C4**  
Gallery: http://192.168.4.1

### Frequency

Ten presets, CH0 (921.47 MHz) to CH9 (930.50 MHz); each tap moves to the next. The gateway asks the node to switch, then confirms on the new channel; without the node's answer both sides return to the previous channel.

### Freq Reset

For when the two sides no longer share a channel, e.g. after a re-flash or an interrupted change. The gateway probes every preset in turn; the node that answers is walked back to CH0.

### LP Freq Reset

The same sweep for a node in low-power standby: a LoRa wake-up preamble goes out on each channel before the probe, so it takes longer.

Freq Reset armed (SURE?)

| 1                                                    | 2                                                              | 3                                                        | 4                                                          | 5                                                           |
|------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| <b>Arm</b><br>Tap Freq Reset; the badge shows SURE?. | <b>Confirm</b><br>Tap again within 5 s; any other tap cancels. | <b>Sweep</b><br>Title shows scan CHn/10 for each preset. | <b>Handshake</b><br>The node that answers is moved to CH0. | <b>Result</b><br>OK, or NO NODE; gateway on CH0 either way. |

### Title Bar States

|                                                                              |                                                                                         |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>CONFIRM · Tap again to reset</b><br>Armed for 5 s; any other tap cancels. | <b>WAIT · Reset: scan CHn/10</b><br>The sweep is running; wait for the result.          |
| <b>OK</b><br>Node confirmed; both sides on the shown channel.                | <b>NO NODE · Reset done, no node</b><br>No node answered; the gateway is on CH0 anyway. |
| <b>FAIL</b><br>Not confirmed; gateway back on the previous channel.          | <b>BUSY</b><br>An image transfer or another change is running.                          |

The gateway and the field camera must run the same firmware version. A re-flashed node starts on CH0: if the gateway shows another channel and captures time out, run Freq Reset once.

## 09 Settings Reference

The interface updates only after a setting succeeds, and the setting is stored on the device.

GW 4.24V

CAM 4.12V

Settings

CFG

CAPTURE

TRIGGER

PIR Motion

Infrared detect

Sound

Mic trigger level

Medium

Timer

Auto-capture interval

5min

AUDIO

Audio Clip

Record 5s with photo

Voice Alarm

Play alert on trigger

Volume

13

SYSTEM

Frequency

Node RF channel

921.47M

Freq Reset

Scan all, back to CH0

RESET

LP Freq Reset

Wake sleeping node first

RESET

Low Power

CAD sleep standby

WiFi: AM36\_B61C4

Gallery: http://192.168.4.1

CAPTURE

Start a remote capture immediately.

PIR Motion

Enable or disable infrared motion triggering.

Sound

Select Off, Low, Med, or High sound triggering.

Timer

Set the automatic capture interval; Off disables it.

Audio Clip

Attach about five seconds of recent pre-capture audio to the image.

Voice Alarm

Play an alert on the field camera after an event image is sent.

Volume

Adjust gateway audio playback volume.

Frequency

Cycle the RF channel (CH0-CH9); it takes effect once the node confirms.

Freq Reset

Scan every channel and bring the node back to CH0; tap twice to confirm.

LP Freq Reset

Same as Freq Reset, but wakes a low-power node on each channel first.

Low Power

Enable low-power standby and the LoRa wake-up path.

Wi-Fi / Gallery

Show the access name and gallery address; tap to enter provisioning or disconnect.

AM36 Settings page (full scroll)

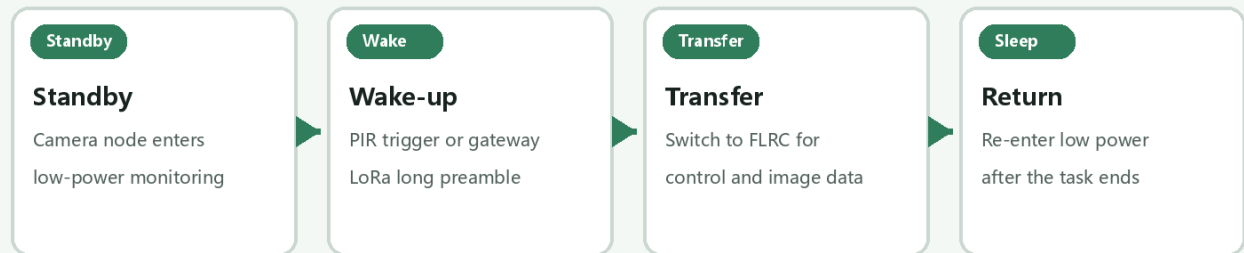
The top of the Settings page shows both gateway and field-camera voltage. The bottom shows the AM36 Wi-Fi name and gallery address. Frequency, Freq Reset, and LP Freq Reset sit in the SYSTEM group further down the page; the previous chapter covers how they work.

## 10 Low-Power Mode

Low-power mode does more than reduce data rate: it switches to a different wake-up and connection path.

### Low-Power Operation

PIR and remote wake-up remain available while low-power mode is enabled.



UI behavior: enabling Low Power turns off and locks Sound and Audio Clip.

| State     | Wireless Operation                                                                                                                  | Available Features                                                              |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Normal    | The field camera continuously receives in FLRC mode; the gateway sends control requests directly and completes the handshake.       | Manual capture, timer, PIR, sound trigger, Audio Clip, and Voice Alarm.         |
| Low Power | The field camera listens for a LoRa wake-up. The gateway first sends a LoRa long preamble, after which the camera switches to FLRC. | Remote capture, timer, PIR, and Voice Alarm. Sound and Audio Clip are disabled. |

### Interface Behavior

|                                                                                                       |                                                                                                                   |                                                                                                           |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Enable Low Power</b><br>Sound returns to Off and Audio Clip is disabled; both controls are dimmed. | <b>Disable Low Power</b><br>Sound and Audio Clip controls become available again but remain off until re-enabled. | <b>PIR Remains Active</b><br>PIR can wake the field camera from low power to capture and return an image. |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|

# 11 Image Reception, Display, and Gallery

From wireless reception to web browsing, the gateway provides a complete result path.

|                                                          |                                                                   |                                                      |                                                                |                                                                |
|----------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|
| <b>1</b>                                                 | <b>2</b>                                                          | <b>3</b>                                             | <b>4</b>                                                       | <b>5</b>                                                       |
| <b>Receive</b>                                           | <b>Verify</b>                                                     | <b>Display</b>                                       | <b>Store</b>                                                   | <b>Browse</b>                                                  |
| The progress page shows fragments, rate, time, and RSSI. | The receiver identifies missing data and requests retransmission. | A complete image is automatically shown full-screen. | The image and optional audio are saved in the gateway gallery. | Open the web page on a phone or computer to view and download. |

GW 4.24V

CAM 4.12V

Receiving

68%

Packets087 / 128

Rate803 kbps

Elapsed00:00.8

RSSI-23 dBm

Progress updates in batches.  
Displayed rate may vary between refreshes.

Image reception progress

GW 4.24V

CAM 4.12V

Link Status

ModeFLRC 2.6M

RSSI-23 dBm

Rate803 kbps

Loss0.5%

Transfer0.8 s

Loss is the first-pass missing-packet ratio.

Link-status page

HTTP Image Gallery

Connect to the display gateway Wi-Fi and open **http://192.168.4.1** in a browser. Images can be viewed and downloaded; records marked AUDIO can play the captured audio clip.

Gallery Limits

Up to 20 records and 6 MB of total storage.

| Page               | Displayed Information                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------|
| Reception Progress | Received fragment count, real-time rate, transfer time, and RSSI.                                      |
| Link Status        | FLRC mode, RSSI, rate, initial missing-data ratio, and total transfer time.                            |
| Web Gallery        | Up to 20 records and 6 MB total storage, with image viewing, downloading, and captured-audio playback. |